



CESAB
CENTRE DE SYNTHÈSE ET D'ANALYSE
SUR LA BIODIVERSITÉ

Why synthesise evidence?

An introduction to systematic reviews,
systematic maps and meta-analyses

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SOUTIENS INSTITUTIONNELS



LVMH





Learning objectives

By the end of this session, you will be able to:

- 1 Explain **why a single study is not enough** to answer an applied question.
- 2 **Distinguish** a narrative review, a systematic map, a systematic review and a meta-analysis — and say what each can and cannot tell you.
- 3 Name the **standards** that make a synthesis trustworthy (CEE, Cochrane, PRISMA 2020, ROSES).
- 4 Place **each step of this week** in the workflow of a systematic review.

Look closer, then step back

One study...

What is this?

One can. Clear, precise, detailed.



Many studies...

And now?

Many cans. Each one looks different.



...and the big picture

Georges Seurat's *A Sunday Afternoon on the Island of La Grande Jatte*, rebuilt from **106,000 cans**.

One study = one can.

A synthesis reveals a pattern that no single study shows — but it can also **hide the details** of each can.



Chris Jordan, *Cans Seurat* (2007), series *Running the Numbers*: 106,000 aluminium cans, the number used in the US every 30 seconds.

Our question: one study, or all of them?

In medicine: one study, twelve children, one retraction

Year	What happened
1998	Wakefield et al. (<i>The Lancet</i>): case series of 12 children , suggests a link between the MMR vaccine and autism. Press conference, huge media coverage.
1998–2004	MMR vaccination coverage falls in the UK; measles returns.
2004	10 co-authors retract the paper's interpretation.
2010–11	<i>The Lancet</i> fully retracts the paper; Wakefield is struck off; the <i>BMJ</i> shows the data were fraudulent .
2014	Meta-analysis of 5 cohort (1.26 million children) and 5 case-control studies: no association between vaccination and autism.

A single, vivid study

small sample, no control group, conflicts of interest — but enormous influence.

The weight of evidence

all eligible studies, weighted by their precision, appraised for bias.

Source: Wakefield et al. (1998); Godlee et al. (2011); Taylor et al. (2014) (cohorts: 1,256,407 children; case-control: 9,920).

Our running example this week

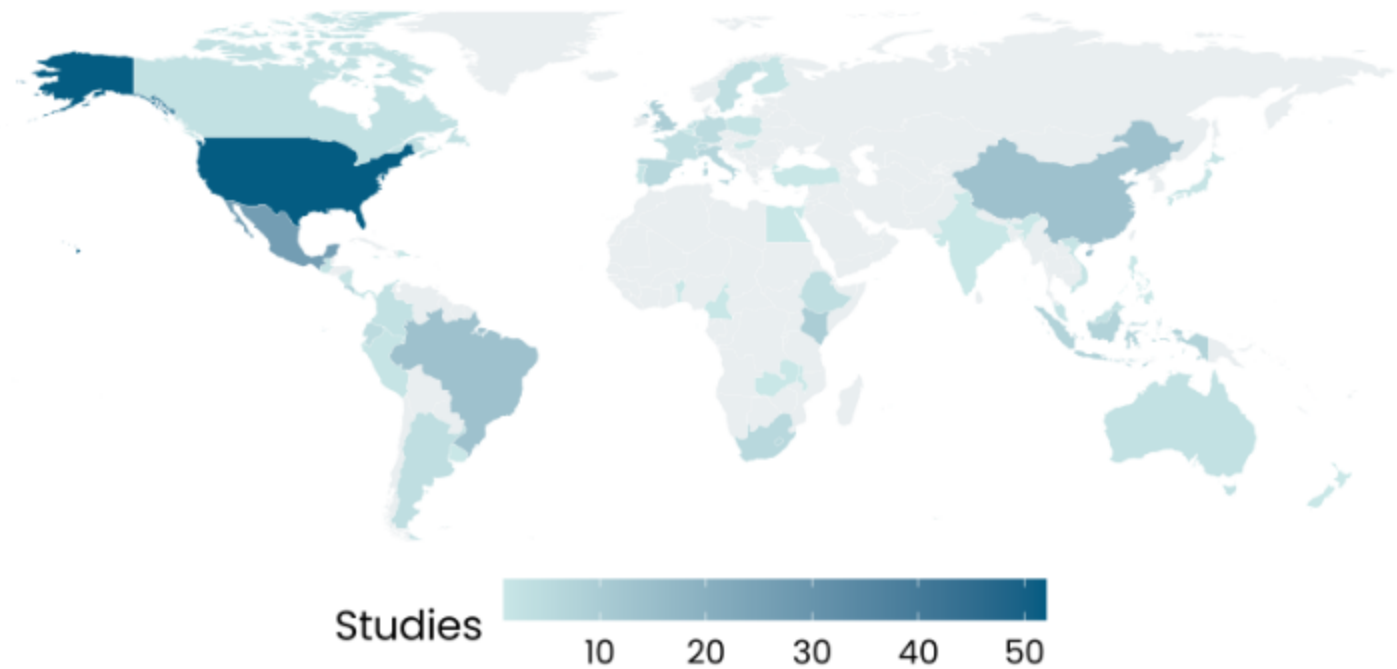
The question

Does **diversifying farms** (intercropping, agroforestry, rotations...) benefit **biodiversity** — without costing **yield**?

A published systematic map (Jones et al. 2021) (partly built from earlier meta-analyses):

- **1,590** records screened
- **237** studies · **4,076** comparisons
- **48** countries · 1986–2021

We use it **all week**: map, effect sizes, models, bias.



Quick poll

SHOW OF HANDS

Compared with a monoculture, does intercropping reduce the abundance of crop pests?

Yes, clearly

No

It depends

If you had read only one study...

● Fewer pests (significant) ● More pests (significant) ● No significant effect

Straub et al. 2013

-90% -75% -50% 0% +100%

Pest abundance, intercropping vs monoculture

Straub et al. 2013: intercropping **increases** pest abundance (+109%).
Conclusion: avoid intercropping?

...or another one

● Fewer pests (significant) ● More pests (significant) ● No significant effect

Straub et al. 2013



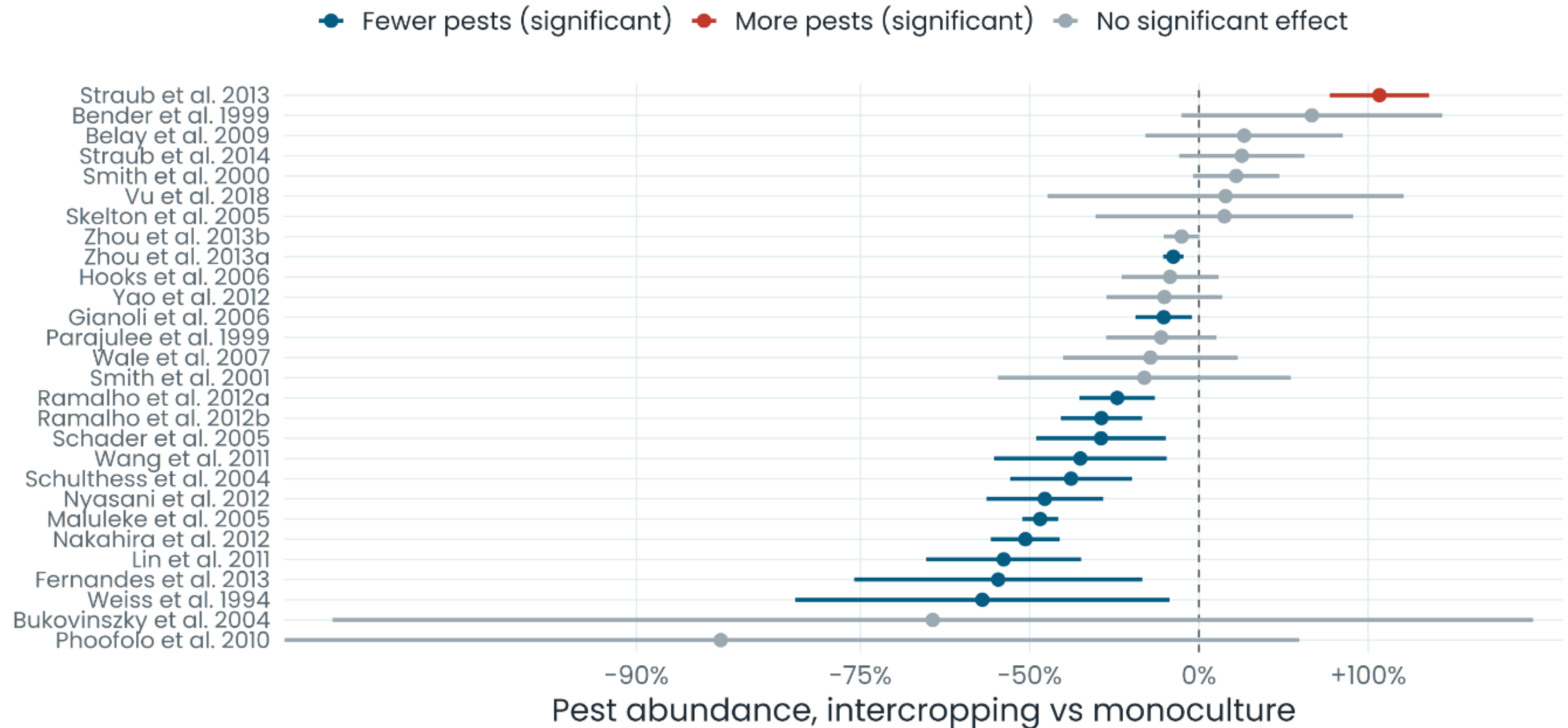
Phoofolo et al. 2010



Phoofolo et al. 2010: intercropping **cuts** pests (-86%). Conclusion: intercrop everywhere?

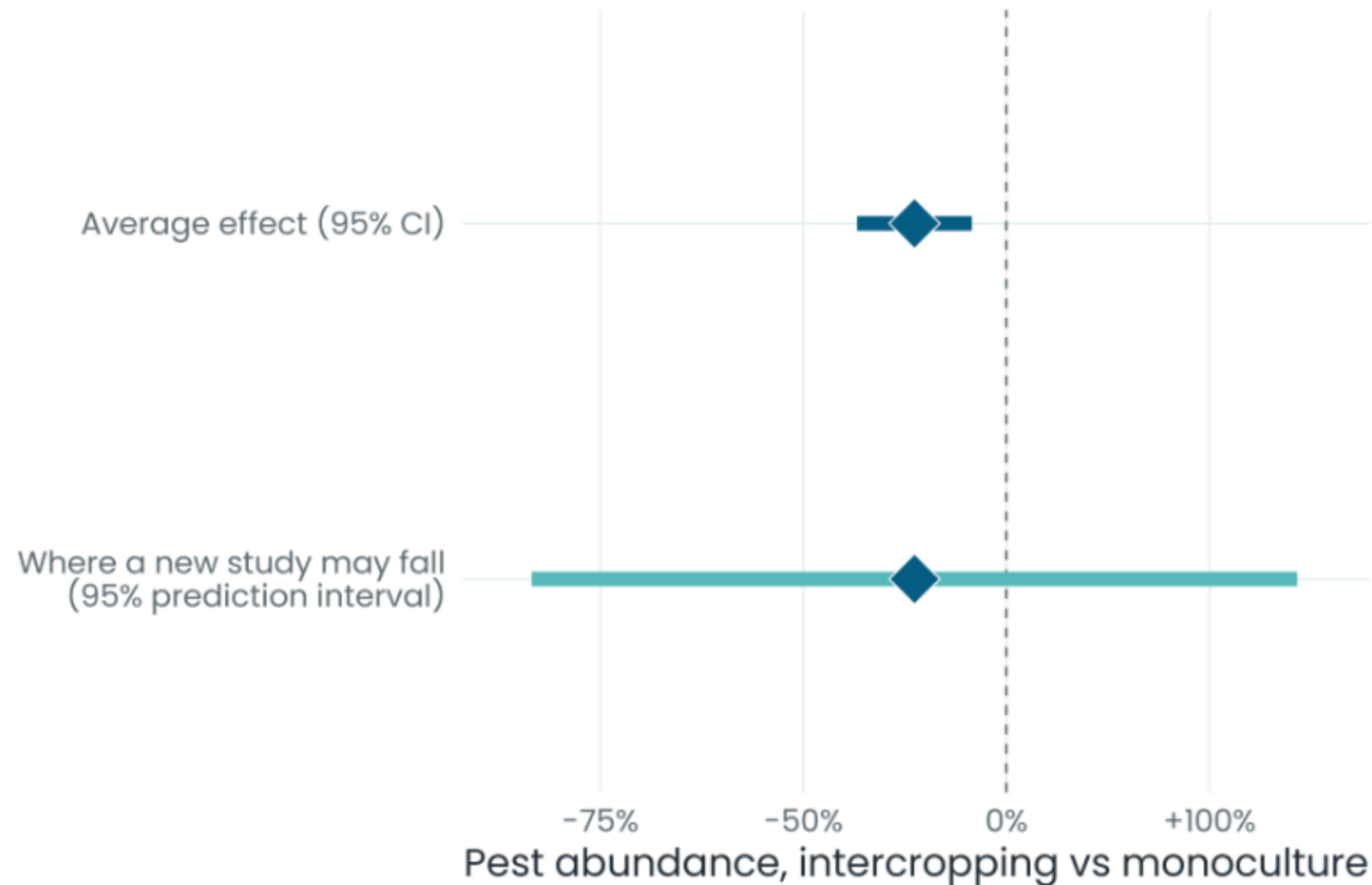
-90% -75% -50% 0% +100%
Pest abundance, intercropping vs monoculture

Primary studies disagree



Source: 28 studies, 382 comparisons from Jones et al. (2021). One point per study (random-effects summary of its comparisons), 95% CI.

A synthesis sees the pattern — and its spread



On average, intercropping **reduces pests by 27%** (95% CI: 11–40% reduction).

But in a new field, the effect could be anywhere from **-80%** to **+170%**.
→ The honest answer to the poll was **“it depends”**.

Random-effects meta-analysis accounting for several comparisons per study: $k = 382$ comparisons, 28 studies. You will fit this model on Wednesday.

Two intervals, two questions

Confidence interval (CI)

*How precisely do we know the **average** effect?*

-40% to -11% → shrinks as studies accumulate.

Prediction interval (PI)

*What effect should we expect in a **new field**?*

-80% to +170% → does **not** shrink to zero with more studies: it reflects **real differences** between sites, crops and pests.

Heterogeneity (τ^2)

The **spread of the true effects** (between studies, and between comparisons within studies). Here the true effect differs from field to field — which is why the PI is wide.

In ecology this is the rule, not the exception (Senior et al. 2016).

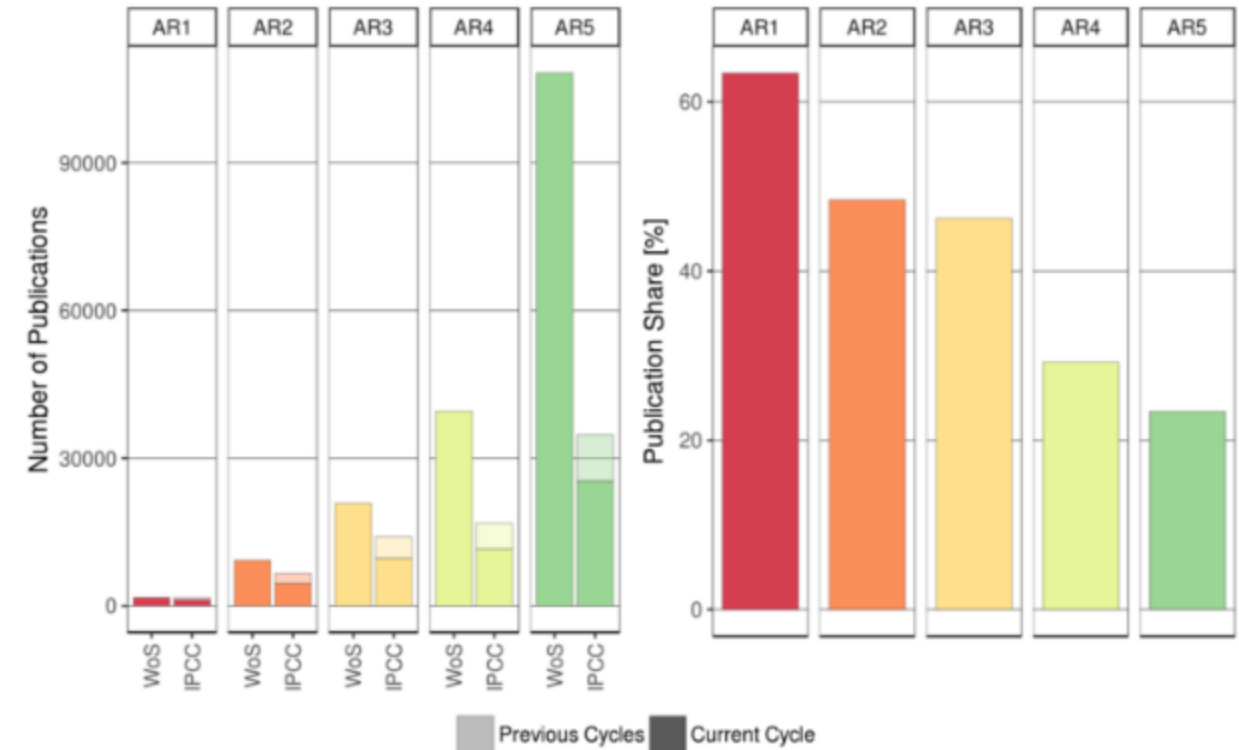
Report the mean **with** its prediction interval. Explaining the spread (crop, pest group, region) is the job of **moderators** — Wednesday afternoon.

Why we need synthesis

Too much to read



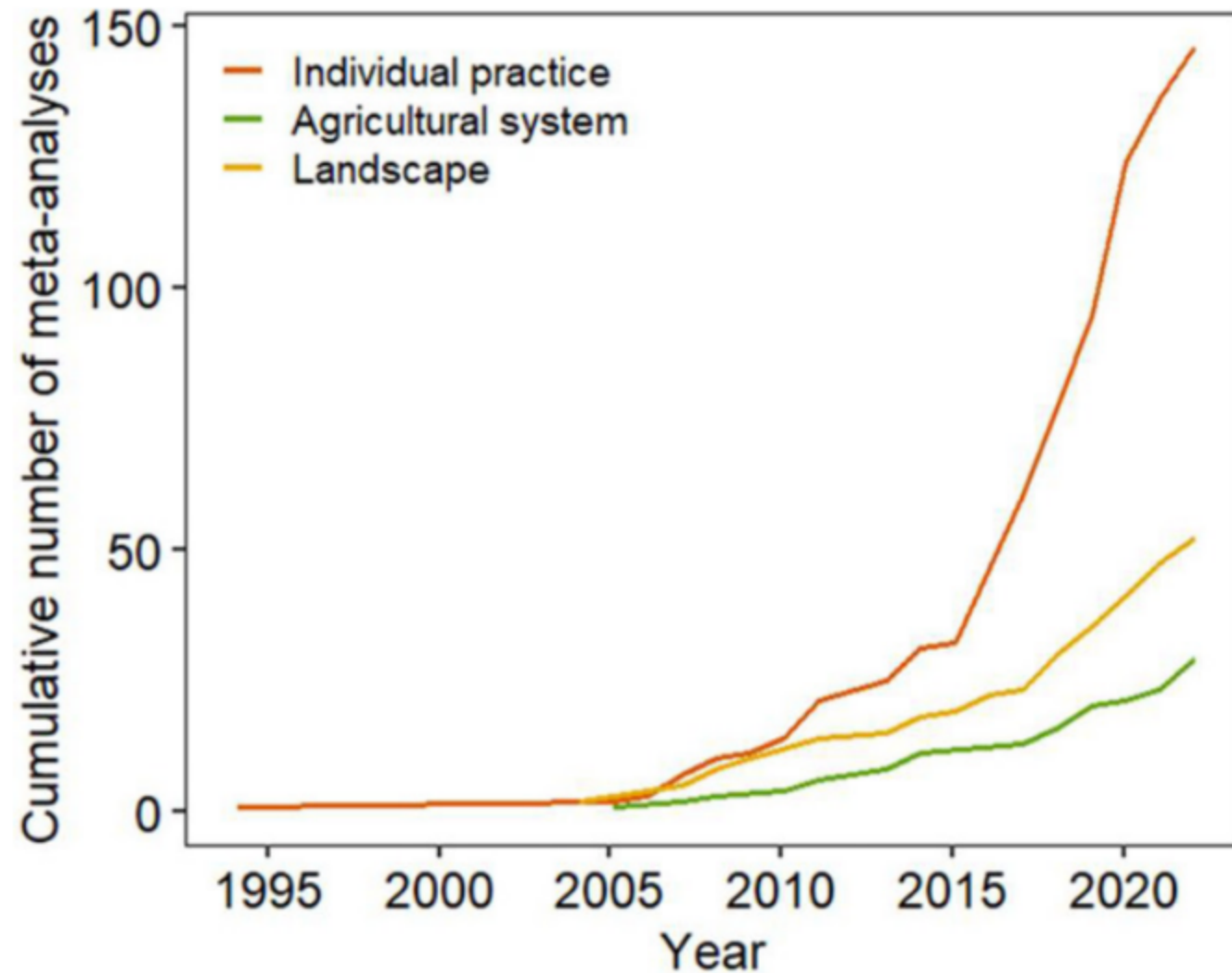
Publications grow exponentially — today > **3 million** peer-reviewed articles per year across all sciences (Johnson et al. 2018).



IPCC assessments cite a **shrinking share** of the relevant literature at each cycle.

Left: Gurevitch et al. (2018). Right: Minx et al. (2017).

Syntheses boom...

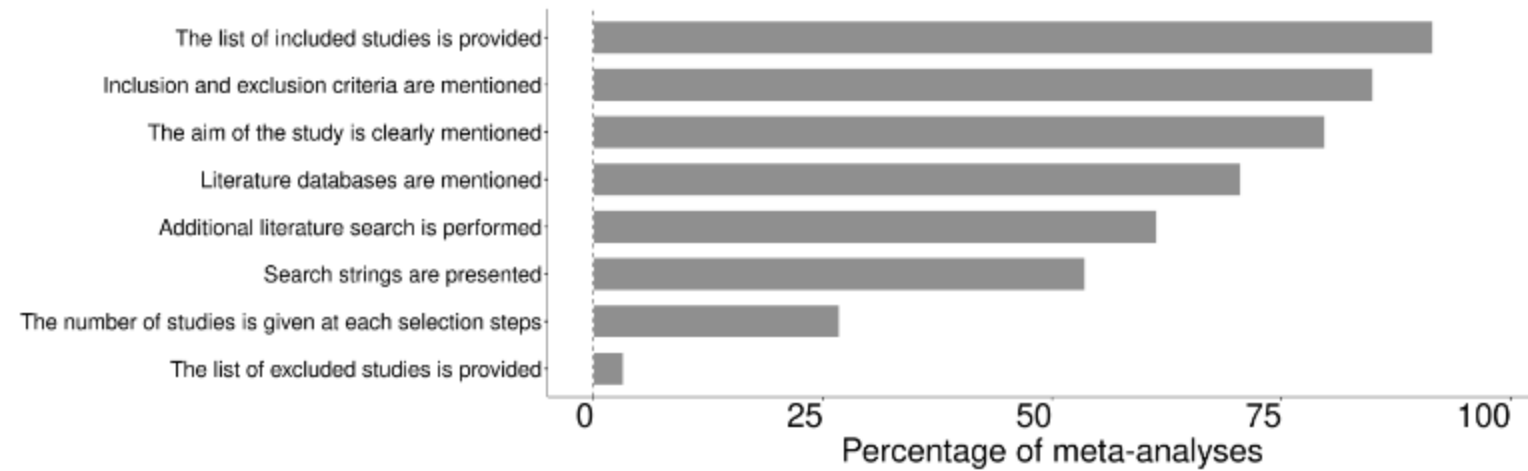


Cumulative number of meta-analyses on **crop diversification**, by scale of the practice.

Dozens of meta-analyses on crop diversification...
Do they agree? Can we trust them?

Source: Beillouin et al. (2019) and updates.

...but their quality varies



Share of meta-analyses on crop diversification reporting each item.

Reporting standards

PRISMA 2020 (Page et al. 2021)

PRISMA-EcoEvo (O'Dea et al. 2021)

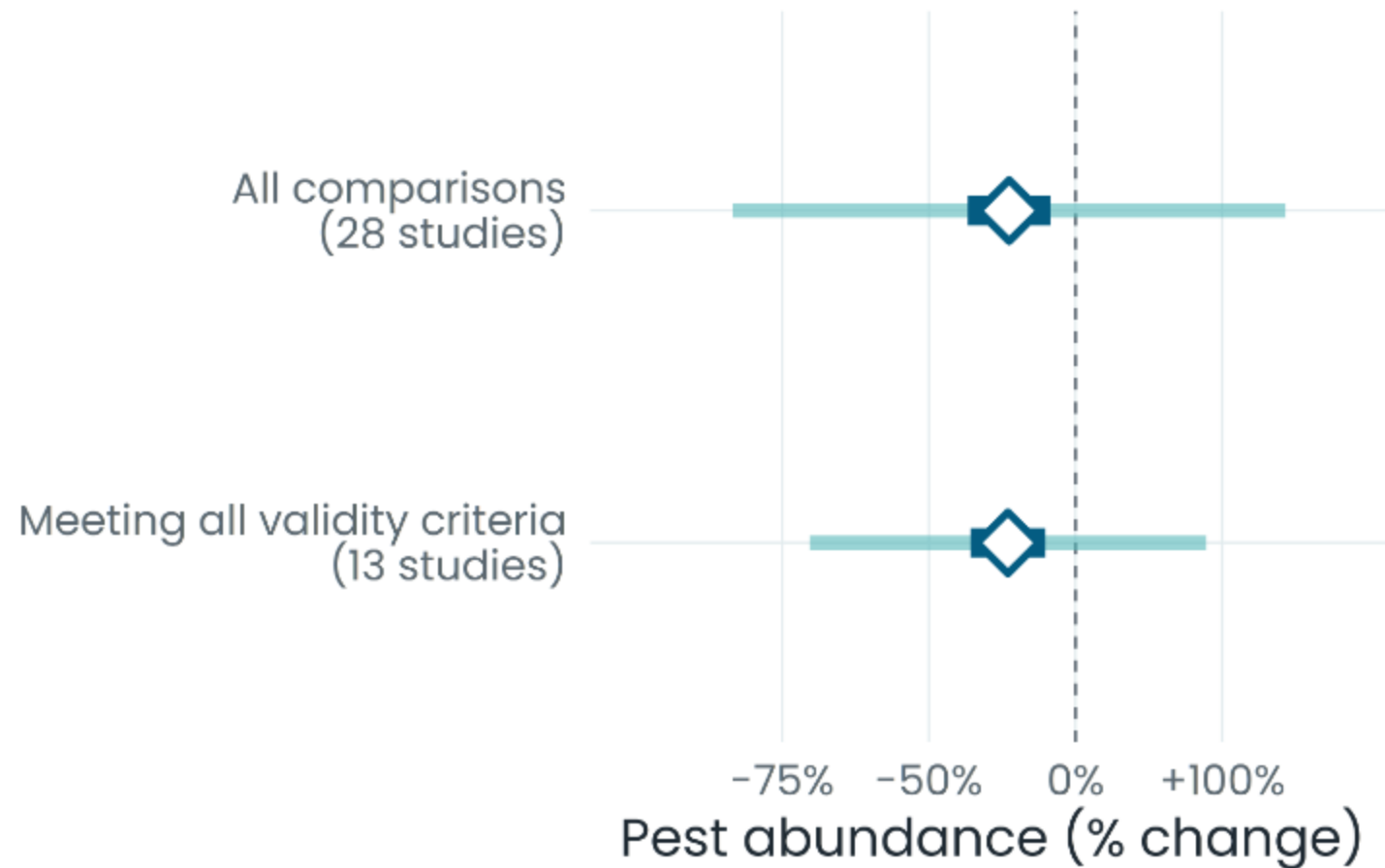
ROSES (Haddaway et al. 2018)

Appraising a synthesis

CEESAT (Woodcock et al. 2014)

Ten appraisal questions (Nakagawa et al. 2017)

Garbage in, garbage out: appraise before you pool



The authors of the map **appraised the comparisons** (replication, duration, distance between plots) – 130 of 382 could not be appraised.

Dark bar = 95% CI, light bar = 95% PI. Restricted to valid comparisons: **same mean** (-27%) and a narrower prediction interval – weaker designs may add noise (or the 13 remaining studies are simply more alike). Appraisal lets you **check** this.

You only know this because validity was **coded**. Reporting standards (PRISMA, ROSES) $\neq$ **critical appraisal** of each primary study – Wednesday morning.

A jungle of terms

Your turn

30 SECONDS · DISCUSS WITH YOUR NEIGHBOUR

**Review, systematic review, systematic map,
meta-analysis: what is the difference?**

Describing the evidence

Method	Question	What makes it distinctive	Output
Narrative review	Broad	Expert selection of studies; no explicit, reproducible method	Expert overview — prone to selection bias
Scoping review	Broad, exploratory	Maps concepts and types of evidence; usually no appraisal	Clarifies a field before a full review
Systematic map	Broad: <i>what is known?</i>	Systematic search and coding of the whole evidence base	Database + map of evidence and gaps

Source: Collaboration for Environmental Evidence (2022); Grant and Booth (2009); James et al. (2016).

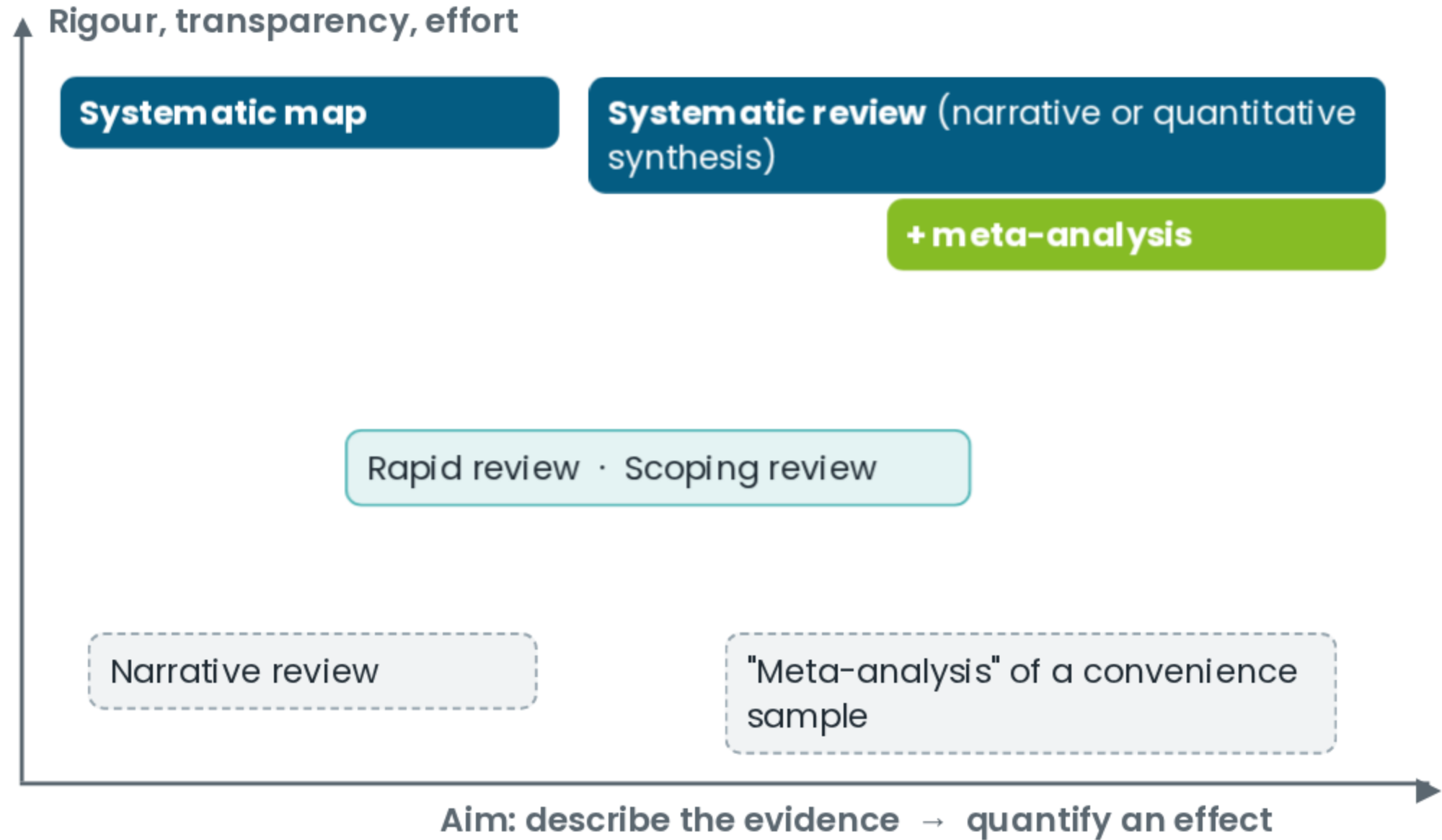
Answering a question

Method	Question	What makes it distinctive	Output
Rapid review	Focused, urgent	Systematic review with <i>declared</i> shortcuts (fewer databases, one screener)	Timely answer, lower certainty
Systematic review	Focused (PICO/PECO)	<i>A priori</i> protocol, comprehensive search, explicit eligibility, critical appraisal , transparent synthesis	Answer to the question, with its certainty
Meta-analysis	Quantitative	<i>Statistical</i> combination of effect sizes, weighted by precision	Mean effect, heterogeneity, moderators

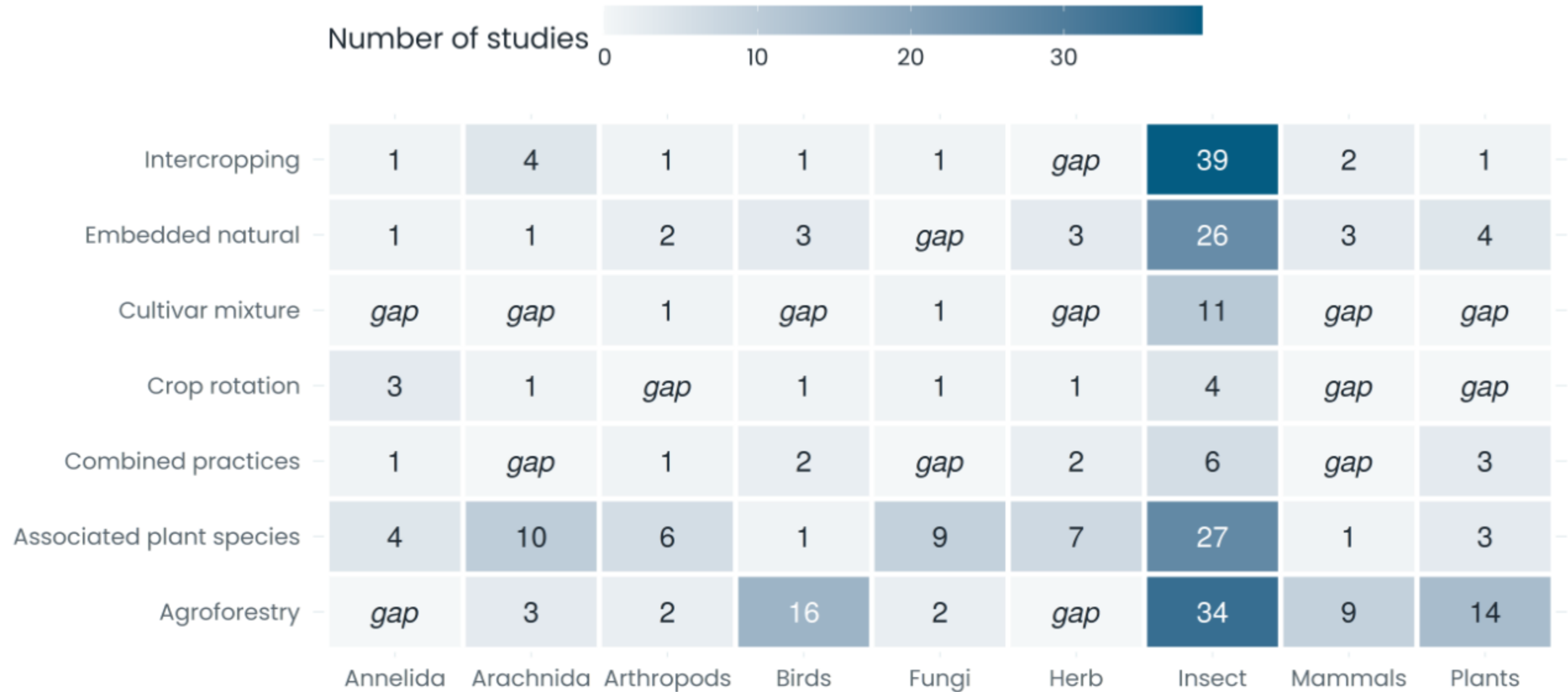
Systematic review $\neq$ meta-analysis. “Systematic” describes the **method**; meta-analysis is a **statistical tool**. Gold standard: a meta-analysis **within** a systematic review.

Source: Collaboration for Environmental Evidence (2022); Higgins et al. (2024). Umbrella reviews apply the same logic to existing reviews.

Choosing a method



A systematic map: where is the evidence?

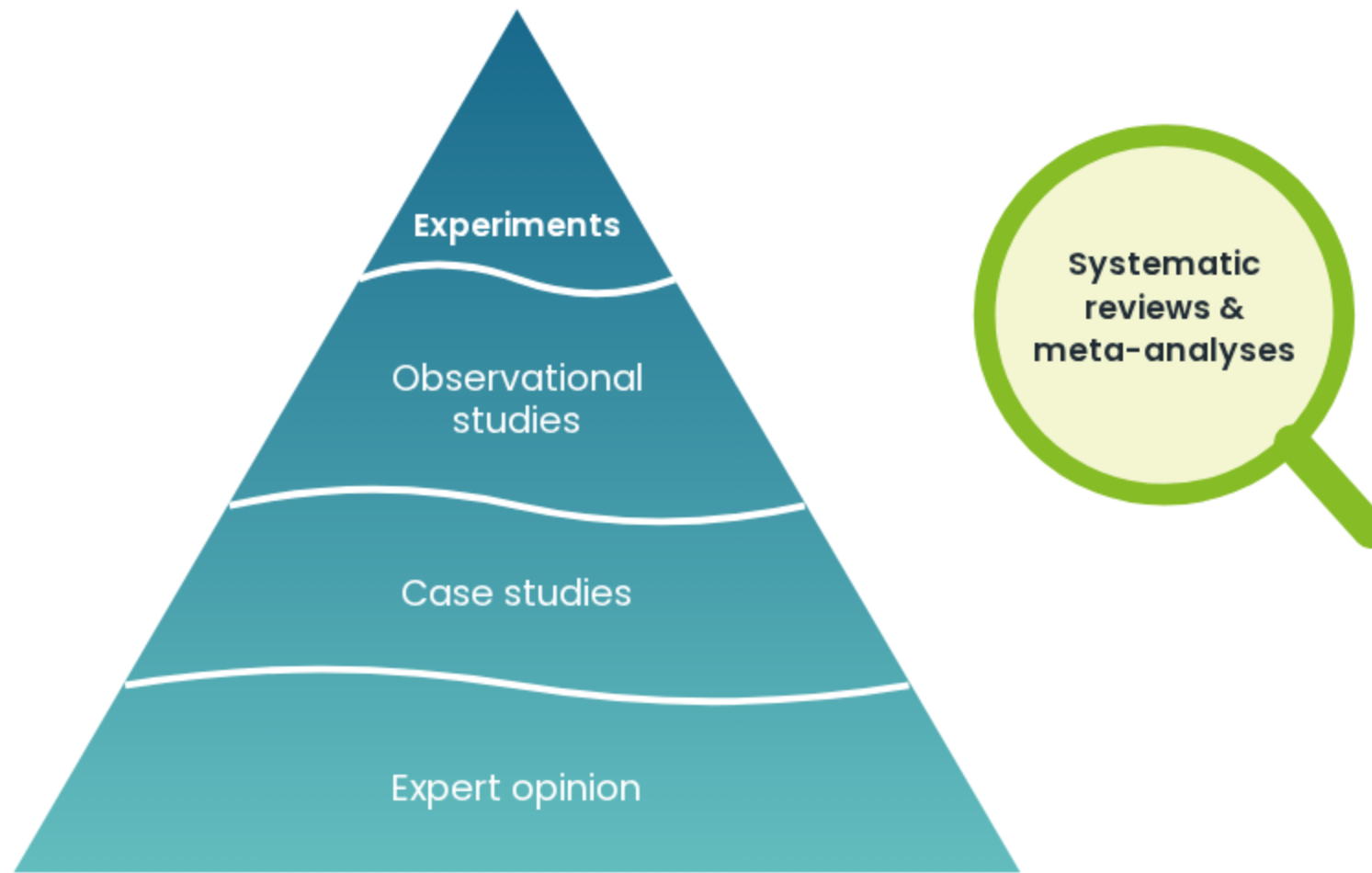


Number of studies per practice × taxon, from Jones et al. (2021). You will build this kind of map on Tuesday afternoon.

Strengths and limits of meta-analysis

Meta-analysis can...	...but it cannot
See the forest for the trees	Show every tree: the average can mask contrasting effects
Quantify heterogeneity (τ^2 , prediction interval)	Make incomparable studies comparable: that is decided by the question
Estimate the average effect of a practice across the studied sites (when studies are experiments)	Turn moderator comparisons (between studies) into causal evidence
Explore small-study effects and test robustness	Remove bias from primary studies: <i>garbage in, garbage out</i>

The evidence pyramid, revisited



The classic pyramid puts meta-analyses **on top**. The revised pyramid (Murad et al. 2016):

- Syntheses are a **lens** through which primary evidence is viewed – not a level above it.
- Boundaries are **wavy**: a well-run observational study can beat a flawed experiment.

A synthesis is only as strong as the studies it contains **and** the method used to find and appraise them.

Standards exist — we follow them this week

Guidelines

CEE Guidelines and Standards, v5.1
(Collaboration for Environmental Evidence
2022) — environment

Cochrane Handbook, v6.5 (Higgins et al. 2024)
— health

Campbell Collaboration — social sciences

Environment: mostly field experiments and observational
studies, strong spatio-temporal variability → expect
heterogeneity.

Reporting

ROSES (Haddaway et al. 2018)

PRISMA 2020 (Page et al. 2021) and PRISMA-
EcoEvo (O'Dea et al. 2021)

Reference books

Koricheva et al. (2013) · Borenstein et al.
(2021)

Harrer et al. (2021) (free online, R code)

Protocol registration

PROCEED (environment) · PROSPERO (health) ·
OSF

Which method for which question?

A. A ministry needs, **within 6 weeks**, an overview of whether hedgerows benefit pollinators.

Rapid review — with its shortcuts declared.

B. Which diversification practices, taxa and regions **have been studied**, and where are the gaps?

Systematic map.

C. **How much** does intercropping change pest abundance, and **why** does it vary?

Systematic review with meta-analysis (and moderators).

Your week: one systematic review, step by step

Mon

Question & stakeholders
Protocol
Search · Screening
Full texts · Reporting

Joseph · Sylvie · Nicolas

Tue

Data visualisation
From hypothesis to map
Codebook
**R practical:
evidence maps**
AI tools

Joseph · Devi · Damien

Wed

Effect sizes · appraisal
Quantitative extraction
Models · bias
Advanced methods

Damien · Joseph

Thu

Group projects:
do it yourself

All

Fri

Presentations
and feedback

All

Workflow: Question → Protocol → Search → Screen → Extract & code → Appraise → Synthesise → Report

KEY MESSAGES

- 1 | **One study can mislead.** Decisions need the whole body of evidence.
- 2 | **“Systematic” is a method** (protocol, exhaustive search, appraisal, transparency); **meta-analysis is a statistical tool.** They are not synonyms.
- 3 | A **map** tells *where* the evidence is; a **meta-analysis** tells *what it says* — and how much it varies.
- 4 | In ecology, **heterogeneity is the rule**: report the prediction interval, not only the mean.

Glossary • English → French

English	Français	English	Français
Evidence synthesis	Synthèse des connaissances	Effect size	Taille d'effet
Systematic review	Revue systématique	Forest plot	Graphique en forêt (<i>forest plot</i>)
Systematic (evidence) map	Carte systématique	Heterogeneity	Hétérogénéité
Scoping review	Revue exploratoire	Confidence interval	Intervalle de confiance
Rapid review	Revue rapide	Prediction interval	Intervalle de prédiction
Meta-analysis	Méta-analyse	Moderator	Modérateur
Critical appraisal	Évaluation critique	Publication bias	Biais de publication
Risk of bias	Risque de biais	Evidence gap	Lacune de connaissances

In the online notebook

Each point of this session developed further, with the same data, more pitfalls and exercises with solutions:

- What is evidence synthesis?
- From research question to estimand
- Designing and running a systematic search
- Screening and study selection
- Data extraction and critical appraisal
- Certainty of evidence and what can be inferred

literaturesynthesis.github.io/notebook

Further reading

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- Borenstein, M., L. V. Hedges, J. P. T. Higgins, and H. R. Rothstein. 2021. *Introduction to meta-analysis*. Second edition. Wiley, Chichester.
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